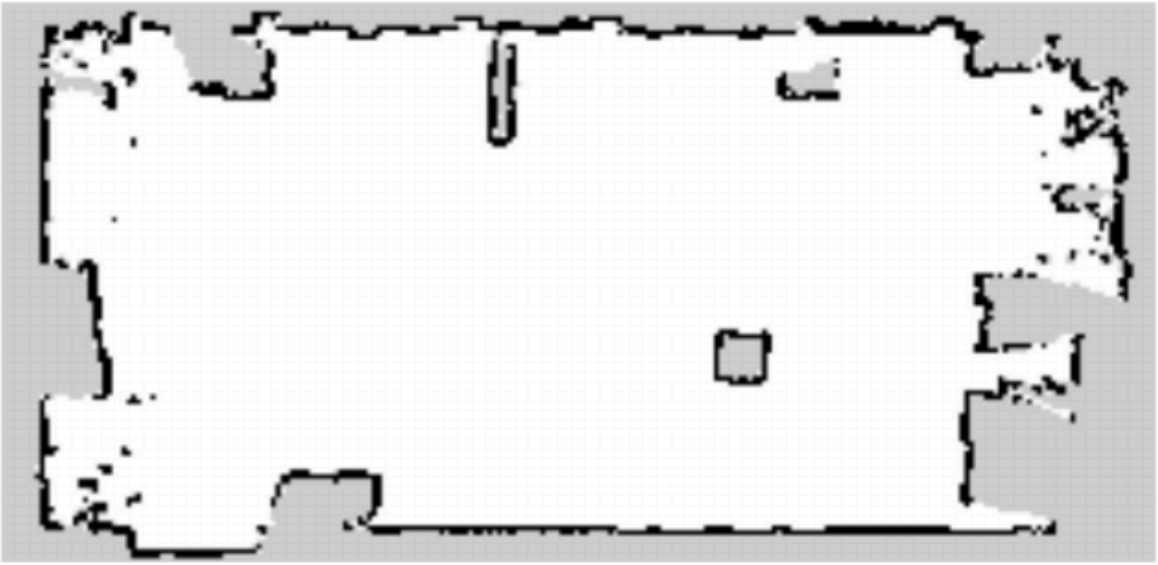
Ro03: Occupancy Grid Maps

- Maps for localisation, path planning, other tasks
- For humans AND robots
- Maps: Representation of Environment
- Building Maps from sensor data

Maps as Sets of Features

- Points and descriptors
- Standard approach when using cameras (CV / photogrammetry)
- **Multiple feature observations improve** 3D position estimate

2D Mapping Example: Occupancy Grid Map



3D Mapping Example: Point Cloud

Mapping as Probabilistic State Estimation

- Compute the map posterior given the sensor data and controls over time

$$p(\underset{map}{m} \mid \underset{control}{u_1}, \underset{observation}{z_1}, \ldots, u_t, z_t)$$

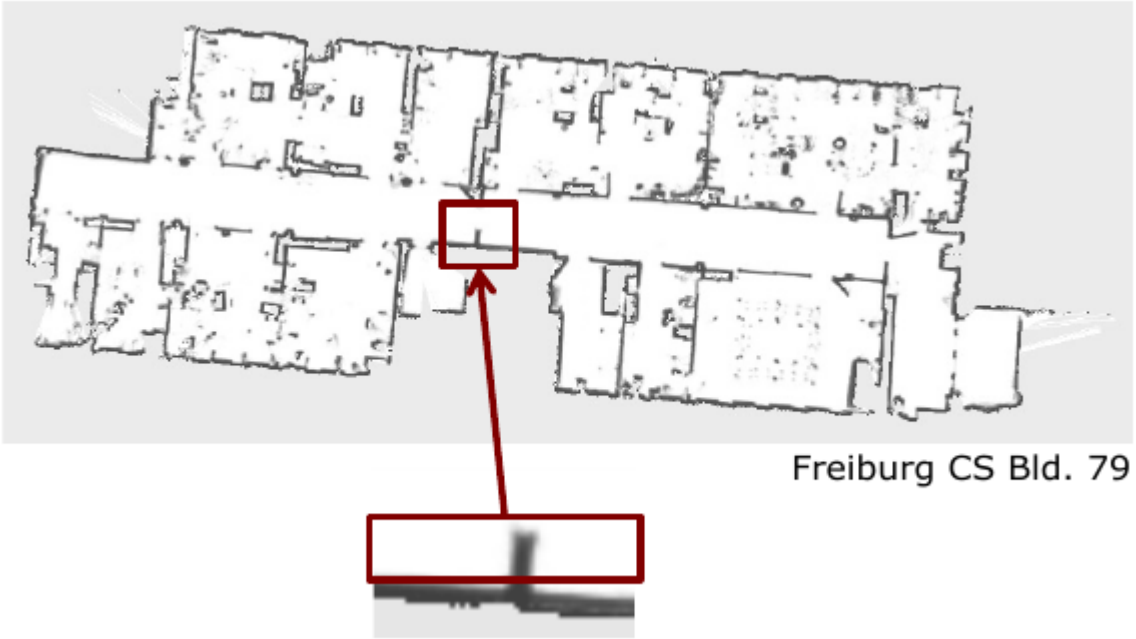
Mapping as Computing the Most Likely Map

- Compute **most likely map** given
  - the sensor data
  - the robots poses ($x_{1:t}$) instead of the sensor controls ($u_{1:t}$)

$$m^* = argmax_m p(\underset{map}{m} \mid \underset{\text{pose instead of control}}{x_1}, \underset{observation}{z_1}, \ldots, u_t, z_t)$$

Occupancy Grid Maps

- Simple Terms
  - Greyscale image
  - Black pixels as occupied space
  - White pixels as free space
- Map to model occupied and free space
- Discretise world via grid
  - Grid structure is rigid
- Each cell assumed to be occupied or free space
- Eliminate unknown states
- **Disadvantage**: Memory (especially 3D)

- No feature detectors needed
- 

Formal Assumptions

1. Cell area either occupied or completely free
   - **Binary Random Variable** modelling occupany of its space
   - $P(M_i = occ) = P_{occ}(M_i) = p(m_i) = 1$
   - $P(M_i = free) = P_{free}(M_i) = 1 - P_{occ}(M_i) = p(\neg m_i) = 0$
   - No knowledge/prior: $p(m_i) = 0.5$
2. World is **static** (Like most mapping systems)
3. Grid cells are **independent** of each other
   - -> $\underset{map}{p(m)} = \prod_i \underset{\text{cell}}{p}(m_i)$

Joint Distributions

$$\underset{map}{p(m)} = p(\underset{cell1}{m_1}, \underset{\text{"and"}}{m_2}, \ldots, \underset{\text{cell N}}{m_N})$$

Example B

- $$\begin{aligned} p\left(M = \begin{smallmatrix} \blacksquare & \square \\ \blacksquare & \square \end{smallmatrix}\right) &= p(M_1 = \blacksquare)p(M_2 = \square) \\ &\quad p(M_3 = \blacksquare)p(M_4 = \square) \\ &= p(M_1 = \blacksquare)(1 - p(M_2 = \blacksquare)) \\ &\quad p(M_3 = \blacksquare)(1 - p(M_4 = \blacksquare)) \end{aligned}$$
- m vs M to distinguish a configuration and the random variable for the map

Map Building: Static State Binary Bayes Filter

- Given sensor date $z_{1:t}$ and the poses $x_{1:t}$ of the sensor, estimate the map:
- $p(m|z_{1:t}, x_{1:t}) = \prod_i p(m_i|z_{1:t}, x_{1:t})$
  - -> binary random variable
- -> Binary Bayes Filter (for a static state)

$$
\begin{aligned}
p(m_i \mid z_{1:t}, x_{1:t}) &\overset{\text{Bayes rule}}{=} \frac{p(z_t \mid m_i, z_{1:t-1}, x_{1:t})\, p(m_i \mid z_{1:t-1}, x_{1:t})}{p(z_t \mid z_{1:t-1}, x_{1:t})} \\
&\overset{\text{Markov}}{=} \frac{p(z_t \mid m_i, x_t)\, p(m_i \mid z_{1:t-1}, x_{1:t-1})}{p(z_t \mid z_{1:t-1}, x_{1:t})} \\
&\overset{\text{Bayes rule}}{=} \frac{p(m_i \mid z_t, x_t)\, p(z_t \mid x_t)\, p(m_i \mid z_{1:t-1}, x_{1:t-1})}{p(m_i \mid x_t)\, p(z_t \mid z_{1:t-1}, x_{1:t})} \\
&\overset{\text{indep.}}{=} \frac{p(m_i \mid z_t, x_t)\, p(z_t \mid x_t)\, p(m_i \mid z_{1:t-1}, x_{1:t-1})}{p(m_i)\, p(z_t \mid z_{1:t-1}, x_{1:t})}
\end{aligned}
$$

Do exactly the same for the opposite event:

$$
p(\neg m_i \mid z_{1:t}, x_{1:t}) \overset{\text{the same}}{=} \frac{p(\neg m_i \mid z_t, x_t)\, p(z_t \mid x_t)\, p(\neg m_i \mid z_{1:t-1}, x_{1:t-1})}{p(\neg m_i)\, p(z_t \mid z_{1:t-1}, x_{1:t})}
$$

By computing the ratio of both probabilities, we obtain:

$$
\begin{aligned}
&\frac{p(m_i \mid z_{1:t}, x_{1:t})}{1 - p(m_i \mid z_{1:t}, x_{1:t})} \\
&= \frac{p(m_i \mid z_t, x_t)\, p(m_i \mid z_{1:t-1}, x_{1:t-1})\, p(\neg m_i)}{p(\neg m_i \mid z_t, x_t)\, p(\neg m_i \mid z_{1:t-1}, x_{1:t-1})\, p(m_i)} \\
&= \underbrace{\frac{p(m_i \mid z_t, x_t)}{1 - p(m_i \mid z_t, x_t)}}_{\text{uses } z_t} \underbrace{\frac{p(m_i \mid z_{1:t-1}, x_{1:t-1})}{1 - p(m_i \mid z_{1:t-1}, x_{1:t-1})}}_{\text{recursive term}} \underbrace{\frac{1 - p(m_i)}{p(m_i)}}_{\text{prior}}
\end{aligned}
$$

From Ratio to Probability

We can easily turn this ratio called the **odds ratio** into the probability:

$$\boxed{Odds(x) = \frac{p(x)}{1-p(x)}}$$

$$p(x) = Odds(x) - Odds(x)\,p(x)$$

$$p(x)\,(1+Odds(x)) = Odds(x)$$

$$\boxed{\begin{aligned} p(x) &= \frac{Odds(x)}{1+Odds(x)} \\ p(x) &= \frac{1}{1+\frac{1}{Odds(x)}} \end{aligned}}$$

$$= \underbrace{\frac{p(m_i \mid z_t, x_t)}{1-p(m_i \mid z_t, x_t)}}_{\text{uses } z_t} \underbrace{\frac{p(m_i \mid z_{1:t-1}, x_{1:t-1})}{1-p(m_i \mid z_{1:t-1}, x_{1:t-1})}}_{\text{recursive term}} \underbrace{\frac{1-p(m_i)}{p(m_i)}}_{\text{prior}}$$

$$= Odds(m_i \mid z_t, x_t)\, Odds(m_i \mid z_{1:t-1}, x_{1:t-1}) \frac{1}{Odds(m_i)}$$

OR

$$= \left[1 + \frac{1-p(m_i \mid z_t, x_t)}{p(m_i \mid z_t, x_t)} \frac{1-p(m_i \mid z_{1:t-1}, x_{1:t-1})}{p(m_i \mid z_{1:t-1}, x_{1:t-1})} \frac{p(m_i)}{1-p(m_i)}\right]^{-1}$$

For reasons of efficiency, one performs the calculations in the log odds notation

Log Odds Notation

The **log-odds notation** computes the logarithm of the ratio of the probabilities (event / not-event):

$$\frac{p(m_i \mid z_{1:t}, x_{1:t})}{1-p(m_i \mid z_{1:t}, x_{1:t})}$$

$$= \underbrace{\frac{p(m_i \mid z_t, x_t)}{1-p(m_i \mid z_t, x_t)}}_{\text{uses } z_t} \underbrace{\frac{p(m_i \mid z_{1:t-1}, x_{1:t-1})}{1-p(m_i \mid z_{1:t-1}, x_{1:t-1})}}_{\text{recursive term}} \underbrace{\frac{1-p(m_i)}{p(m_i)}}_{\text{prior}}$$

$$\Rightarrow \quad l(m_i \mid z_{1:t}, x_{1:t}) = \log\left(\frac{p(m_i \mid z_{1:t}, x_{1:t})}{1-p(m_i \mid z_{1:t}, x_{1:t})}\right)$$

- Log odds ratio defined as
  - $l(x) = \log \frac{p(x)}{1-p(x)}$
- and with p(x)
  - $p(x) = 1 - \frac{1}{1+\exp l(x)}$

Occupancy Mapping in Log Odds Form

- Products turn into a sum

$$l(m_i|z_{1:t}, x_{1:t}) = \underbrace{l(m_i|z_t, x_t)}_{\text{inverse sensor model}} + \underbrace{l(m_i|z_{z:t-1}, x_{1:t-1})}_{\text{recursive term}} - \underbrace{l(m_i)}_{prior}$$

- or in short
  - $l_{t,i} = inv_sensor_model(m_i, x_t, z_t)) + l_{t-1,i} - l_0$

occupancy_grid_mapping$(\{l_{t-1,i}\}, x_t, z_t)$**:**

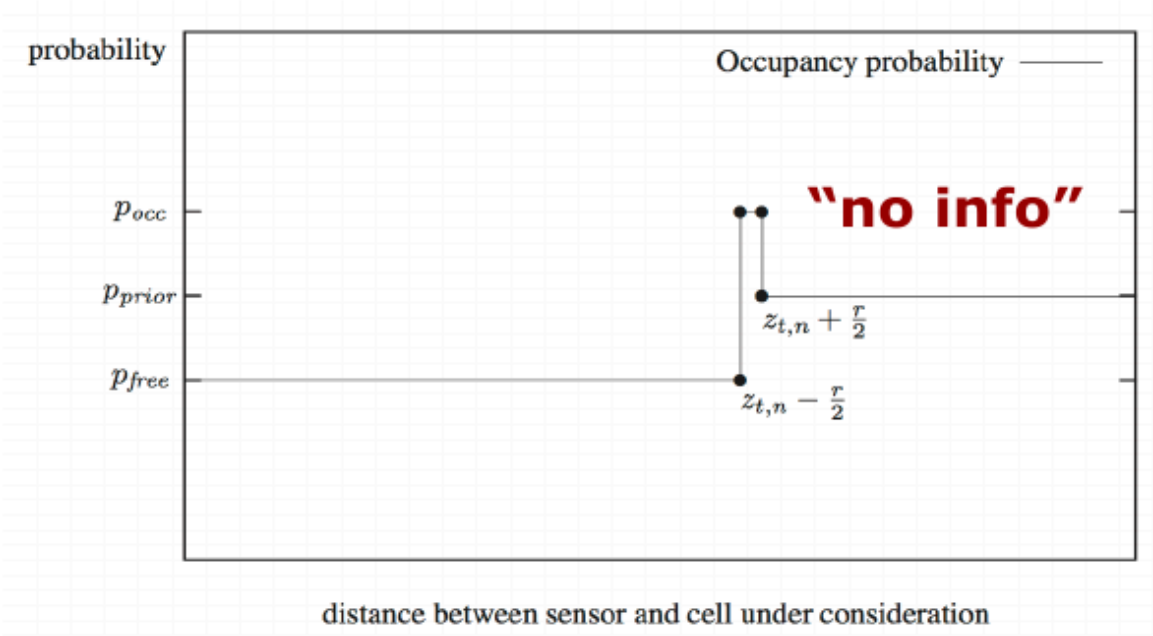
```
for all cells m_i do
    if m_i in perceptual field of z_t then
        l_{t,i} = l_{t-1,i} + inv_sensor_model(m_i, x_t, z_t) - l_0
    else
        l_{t,i} = l_{t-1,i}
    endif
endfor
return {l_{t,i}}
```

highly efficient, we only have to compute sums

- Developed 1985
- **for noisy wide-angle sonar sensors**
- "*Mapping with known poses*"

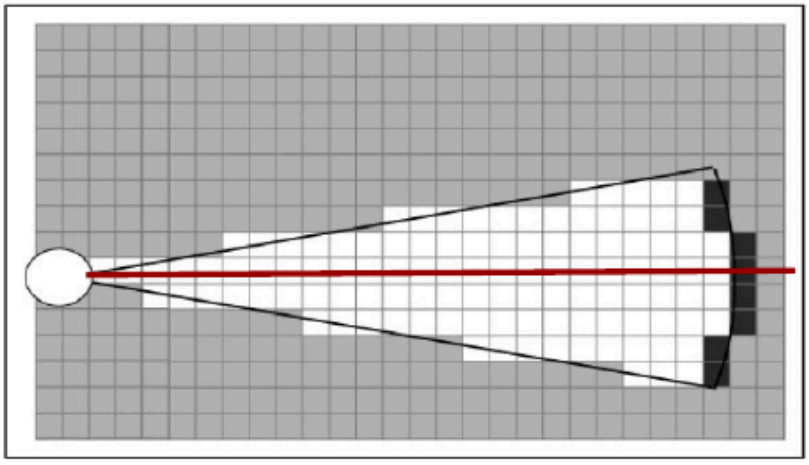
Inverse Sensor Model

Inverse Sensor Model for Precise Range Data (LiDARs)



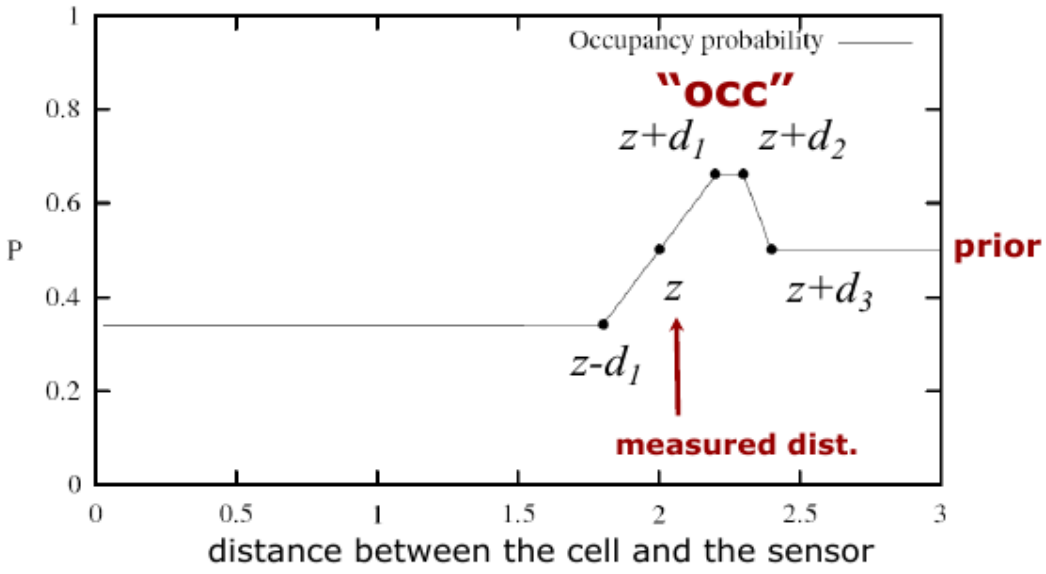
- Bresenham's line algorithm to get relevant cells

Inverse Sensor Model Noisy Range Data Assuming a Sensing Cone (Sonar Sensors)

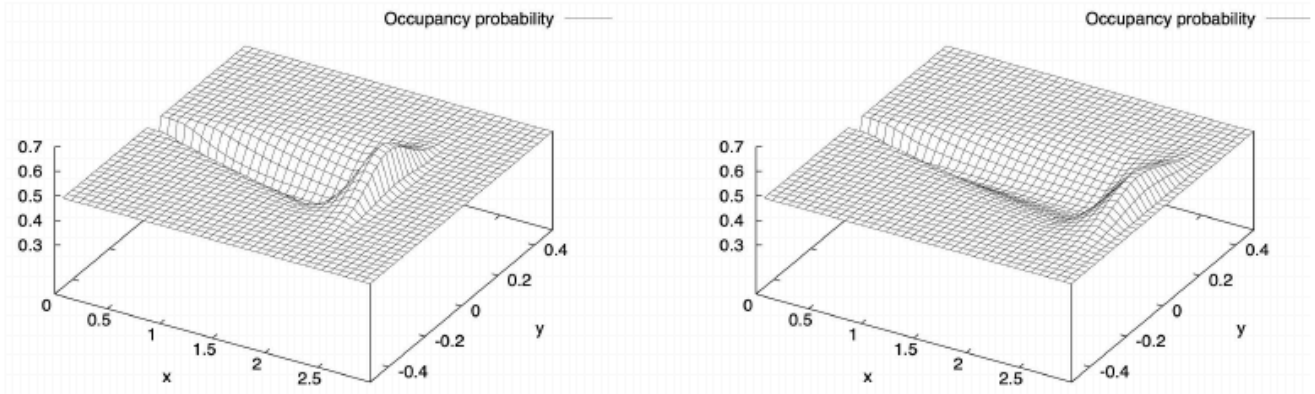


- In the following, we only consider the cells along the optical axis (red line)

Inverse Sensor Model Noisy Range Data (Sonar Range Sensors)

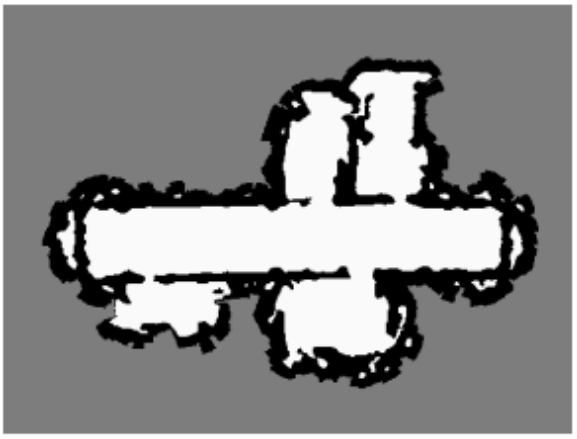


The Impact of the Measurement on the Map Depends on the Measured Distance



2m range reading 2.5m range reading

Resulting Occupancy and Maximum Likelihood Map



The maximum likelihood map is obtained by rounding the probability for each cell to 0 or 1.

Occupancy Grid Map Summary

- Occupancy grid maps **discretise space** into **independent** cells
- Each cell is **binary random variable** (occupancy estimation)
- Implements a **static state binary Bayes filter** per grid cell
- **Log-Odds** representation is **fast** to compute
- So far: **known poses** - mapping with unknown poses is substantially harder ("SLAM")